

IN5340 Project III: Detection theory

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Upload answers to devilry.ifl.uio.no
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Introduction

In this project we will study different aspects of simple hypothesis testing. The problems depends heavily on the theory and examples given in Chapter 4 of *Detection theory, Applications and Digital Signal Processing* [?]. I advice you to study the chapter thoroughly before answering this project.

1 The detector

Assume that want to detect *zeros* and *ones* transmitted over a channel. The *zeros* are coded as no signal while the *ones* are coded as a positive voltage θ volts. The channel introduces additive Gaussian noise, $n \sim \mathcal{N}(0, \sigma^2)$ to the measurements x . The noise is *i.i.d.* (independent and identically distributed).

The problem is illustrated in Figure 1.

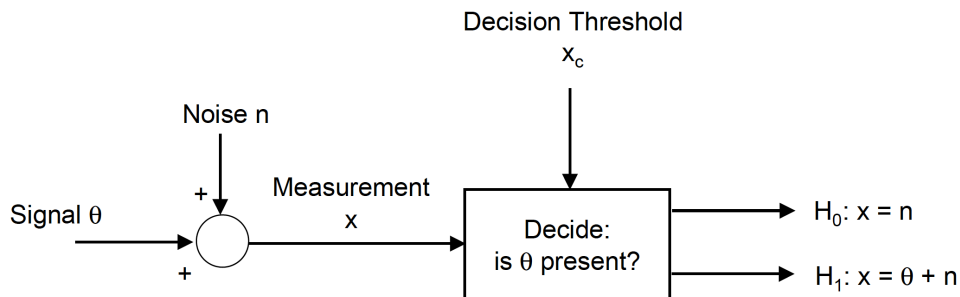


Figure 1: Illustration of the detection problem

We have two hypothesis:

$$H_0: x = n$$

$$H_1: x = \theta + n,$$

where the PDF of the noise is given as $f_n(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}$.

- What is the PDF $f_\theta(x)$ of the *signal + noise*.
- Assume for now that $\theta = 5$ and $\sigma = 5$. Make a plot of $f_n(x)$ and $f_\theta(x)$ (as in illustration 4.5 in [?]). With your own words, how would you overall rate this problem?

c) For the given problem, we may design a CFAR detector with $P_{FA} = 0.1$. If we do so:

- What would then the threshold be?
- How large is P_D in this case?
- Indicate the threshold in the figure made in b).
- In the same figure, do also indicate which area P_{FA} and P_D represent.

d) We now consider a detector where we assume equal probability of receiving *noise* and *signal+noise*. The noise and signal+noise has the PDFs and parameters as above (i.e. $\theta = 5$ and $\sigma = 5$). For this detector, we want experimentally to find the ROC curve of the detector. Construct the "true" and measured data (when using Matlab) as

```
theta = 5;
sigma = 5;
nData = 1000;
s = theta * (randn(nData,1)>=0); % "true" data
x = s + sigma*randn(size(s)); % measured data
```

- Estimate and plot the ROC curve for this detector.
- In the same figure, plot the true (expected) ROC curve.
- Run several realization of the random "true" data and comment on your results.

e) We wish to improve our detector and decide to use more than one measurement when transmitting, and then also detecting *noise* or *signal+noise*, i.e.

$$H'_0: x_n = n_n, \text{ for } n = 1, \dots, N.$$

$$H'_1: x_n = \theta + n_n, \text{ for } n = 1, \dots, N.$$

We choose to use the sample mean as our sufficient statistics.

- Find the joint PDFs of this statistics as a function of the number of measurements used.
- Constructing a CFAR detector, what is the least number of measurements (N) we need if we want $P_{FA} = 0.01$ and $P_D > 0.9$.

f) For the value of N found in e), plot the joint PDFs in the same way as in b) and indicate the threshold found. Comment your results.

g) Again, we wish to consider a problem like the one in d). The noise and signal has the PDFs and parameters as above (i.e. $\theta = 5$ and $\sigma = 5$), but we now use $nMeasures$ to detect each signal. Again, we wish to find the ROC curve of the detector. Construct the "true" and measured data (when using Matlab) as

```
theta = 5;
sigma = 5;
nData = 1000;
signal = theta*(randn(nData,1)>=0); % "true" data, one sample

nMeasures = 4; % No of repetitions
s = repmat(signal.', nMeasures, 1); % Construct matrix containing repeated "true" signals
s = s(:); % Vectorize matrix

x = s + sigma*randn(size(s)); % Add different noise on each component

x = reshape(x, nMeasures, length(signal)).'; % Rearrange into a matrix, again.
```

For a given threshold th , the detected signal, *detector*, can now be found as

```
detektor = theta * (mean(x,2)>=th);
```

- Estimate and plot the ROC curve for this detector for $nMeasures$ being 1, 2, 3, 7, 14.
- In the same figure, plot the true (expected) ROC curves.
- Comment on your results.

- h)** Going back to the simpler case having one sample per detection, we now want to consider a signal with fewer '1' than '0' (i.e. more seldom *signal+noise* compared to only *noise*, but we assume the number of '1' and '0' to be equal. Let $\theta = 5$ and $\sigma = 2.5$. Construct the "true" and measured data (when using Matlab) as

```
theta = 5;
sigma = 2.5;
nData = 1000;
s = theta * (randn(nData,1) >= 1/sqrt(2)); % "true" data
x = s + sigma*randn(size(s)); % measured data
```

- Estimate and plot the ROC curve when assuming having equal number of '1' and '0'.
 - In the same figure, plot the true (expected) ROC curve.
 - Comment on your results.
- i)** We now want to consider detector where we take into account that we know the ratio of '1' and '0'. Repeat the exercise in **h)**, i.e.
- Estimate and plot the ROC curve for this detector.
 - In the same figure, plot the true (expected) ROC curve.
 - What is the consequence of using the detector in **h)** and the benefit of using the detector in **i)**?
 - Comment on your results.

Matlab code

At <https://www.uio.no/studier/emner/matnat/ifi/IN5340/v23/login-protected-info/mcode/> you will find the following matlab functions

Q.m A function that computes the complimentary error function.

Qinv.m A function that computes the inverse complementary error function.

The presentation

Please provide an electronic presentation containing a description of the problem, all needed derivations, results, m-code if needed, and discussion. You should be prepare to present you work, and the presentation should not last more than approximately 15 minutes. Please also provide your matlab source code in addition to a run-time example (if needed) of you code (in Matlab, use 'diary'). If you prefer to write some of it by hand, it is okay. Please then scan the handwritten pages and include it in your presentation.

Upload your solutions in time to <https://devilry.ifi.uio.no>.

If you are using $\LaTeX$, the "funny" $\begin{matrix} H_1 \\ > \\ < \\ H_0 \end{matrix}$ may be written as

```
\begin{subarray}{c}
H_1 \ \vphantom{H_1} \ \backslash\backslash \\
> \ \vphantom{H_1} \ \backslash\backslash \\
< \ \vphantom{H_1} \ \backslash\backslash \\
H_0 \ \vphantom{H_1}
\end{subarray}
```

if you have included the amsmath-package.